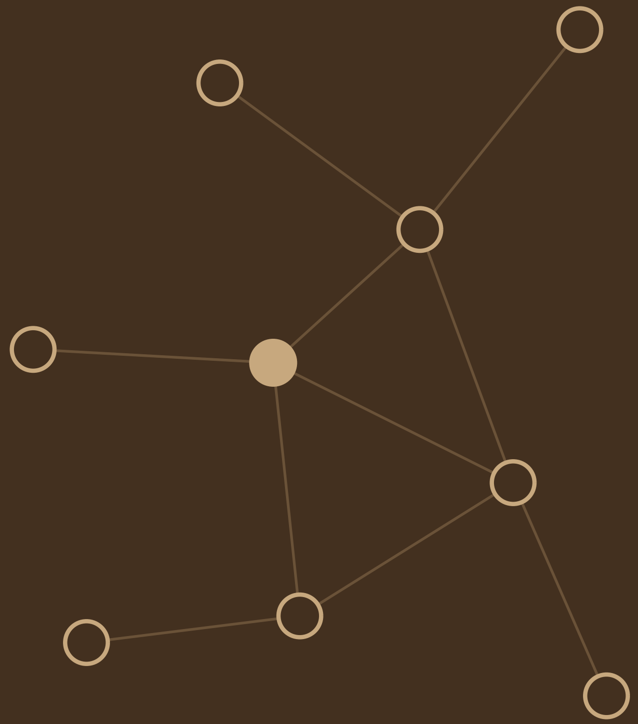
INNOPOLIS UNIVERSITY · BACHELOR'S THESIS · 2026

Secure Communication for Swarm Robotics

Adaptive lightweight cryptography that switches at runtime.



AUTHOR
Georgii Butakov

SUPERVISOR
Oleg Bulichev

THE PROBLEM

No single cipher fits a whole mission

Security and performance are coupled —
balance them at runtime, not at design time.

efficiency *protection*

LIGHTWEIGHT **BALANCED** **HEAVY**

Always heavy

wastes throughput & battery

Always light

unsafe when threat rises

THIS THESIS

Design it formally — then measure it

01

MODEL

Swarm graph + adaptation function

`F : S(t) → P`

02

PROTOTYPE

Pairwise testbed, four modes

`incl. adaptive switching`

03

EVIDENCE

Measured, not just estimated

`228 benchmark runs`

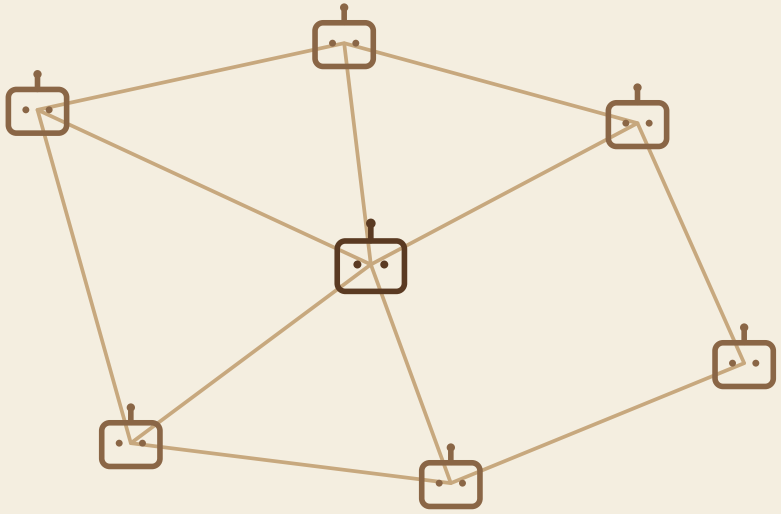
Novelty: a bounded formal model paired with an executable benchmark.

FORMAL MODEL

The swarm as a dynamic graph

G (V, E, t)

nodes = agents · edges = live links



each node carries energy · keys · position

THE ADAPTATION FUNCTION

$$F : S(t) \rightarrow P$$

S(t)

energy · threat · density · loss

↓

F

↓

P

● heavy ● balanced ● light

THREAT MODEL

What attackers do — what we guarantee

THE ADVERSARY CAN

Eavesdrop
read traffic

Tamper
inject / modify

Compromise
leak keys & memory

THE DATA PLANE GUARANTEES

Confidentiality
AEAD, unique nonce

Authenticity
HMAC / hash token

Integrity & containment
AEAD tag · epoch rekeying

BUILDING BLOCKS

One hybrid stack — each primitive where it fits

SESSION SETUP

X25519 + HKDF-SHA256

ephemeral key exchange per epoch → forward secrecy

periodic

PAYLOAD

AES-GCM / ChaCha20-Poly1305

AEAD confidentiality + integrity

per message

RUNTIME AUTH

HMAC-SHA256 / hash token

lightweight origin & replay check

per message

WHY HYBRID

Costly asymmetric work stays off the per-packet path.

PROFILES

Three operating points

HEAVY	BALANCED	LIGHTWEIGHT
AES-256-GCM	AES-192-GCM	ChaCha20-Poly1305
+ HMAC-SHA256	+ HMAC-SHA256	+ hash token
0.250 MiB	0.250 MiB	0.125 MiB
crypto work / message	crypto work / message	crypto work / message

Lightweight does half the per-message crypto work of the authenticated profiles.

ADAPTATION LOGIC

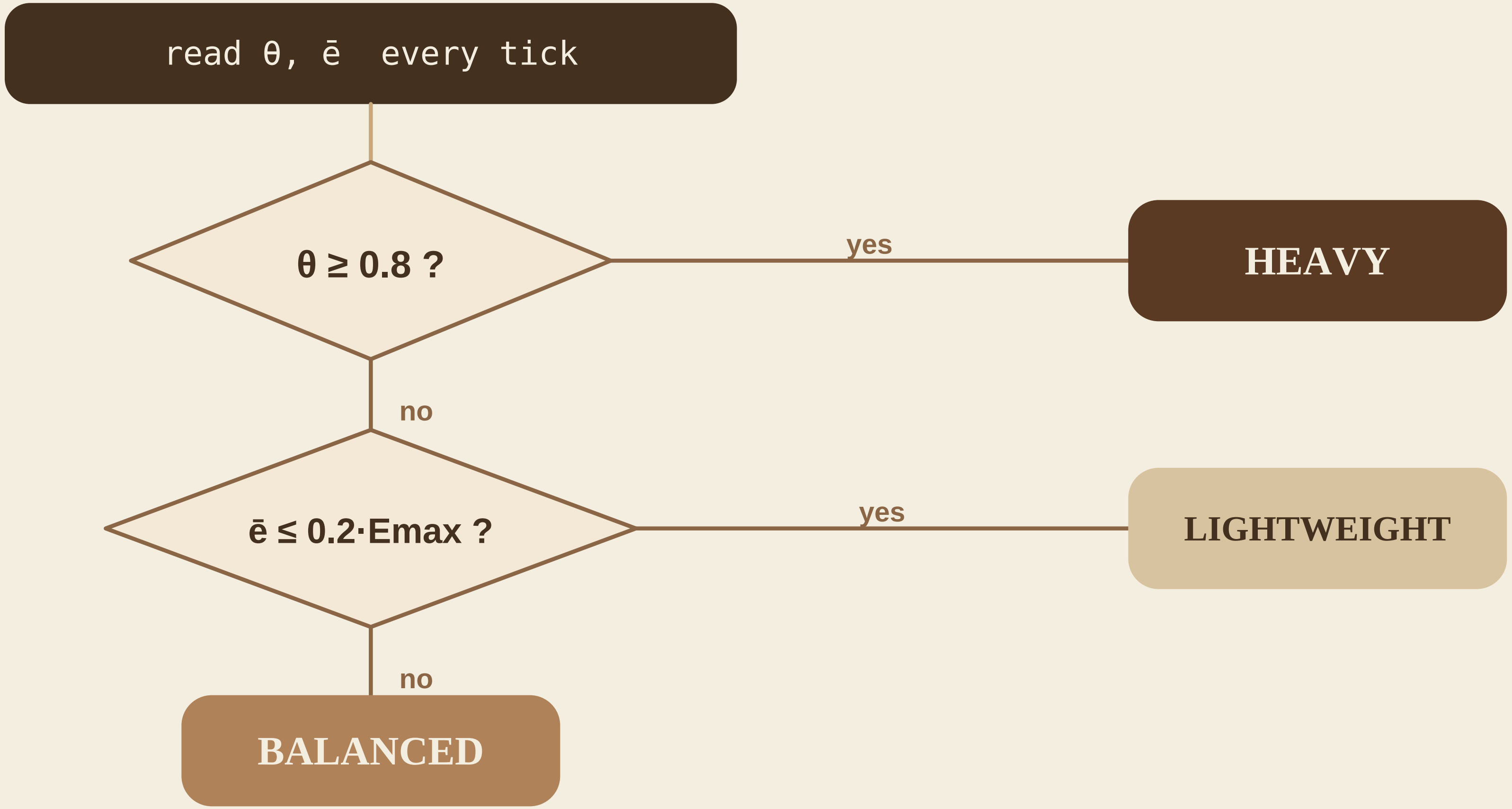
A transparent threshold selector

TWO TRIGGERS

↑ threat
escalate to heavy

↓ energy
fall back to light

otherwise
stay balanced



BENCHMARK

A constrained, reproducible testbed

two containers · encrypted data path

0.05	128	64	60	×3
CPU CORES	MiB RAM	KiB / MSG	SECONDS	REPEATS

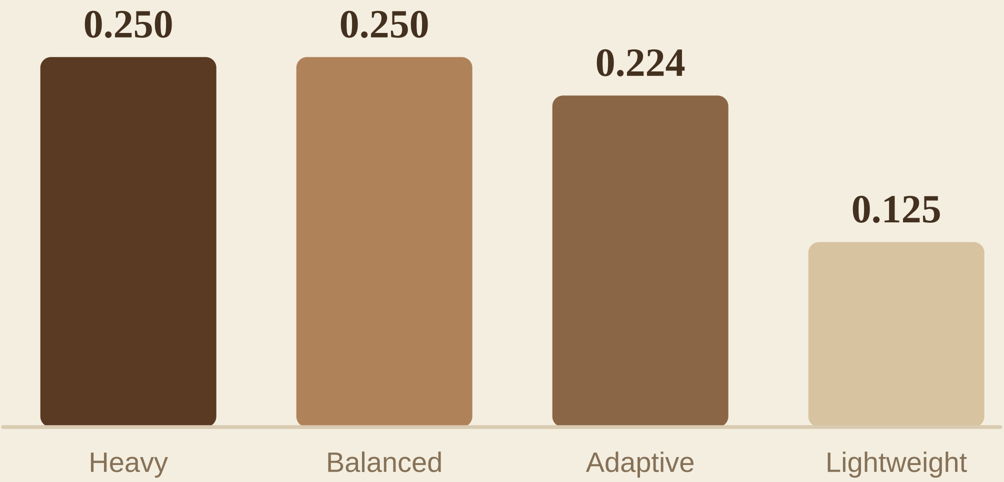
Measured: latency · throughput · drops · switches · crypto work

RESULTS · I

Adaptive spends protection where it is needed

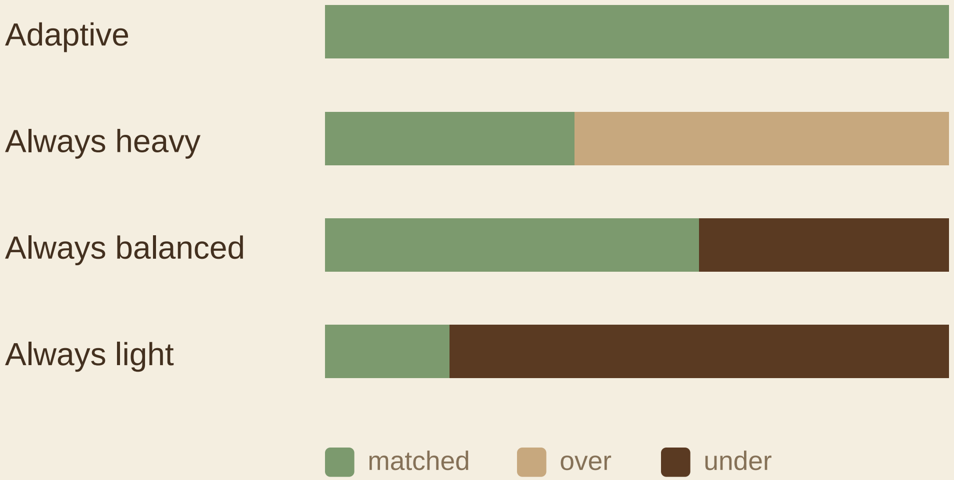
CRYPTO WORK / MESSAGE

MiB · lower is cheaper



POLICY FIT OVER THE 60-s SCHEDULE

share of time at the right protection level



−26.7% **less protection cost**

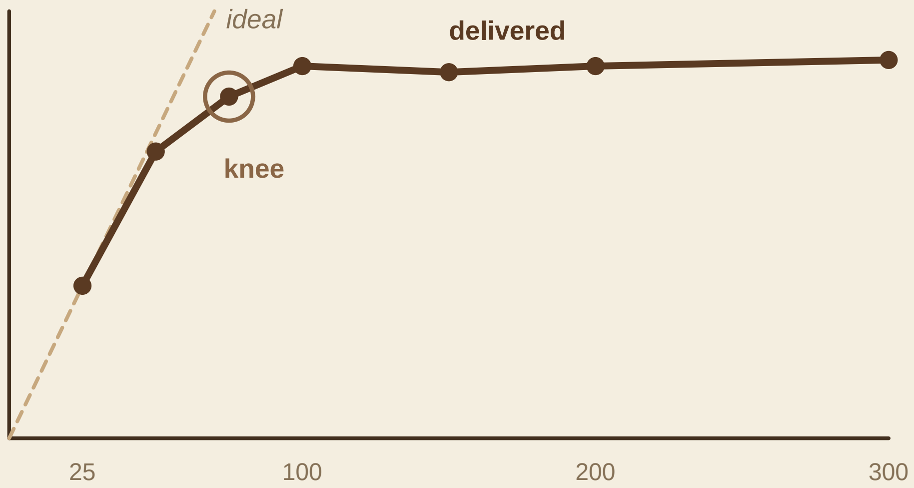
vs. always-heavy · full coverage

RESULTS · II

Under a tight CPU cap, transport dominates

DELIVERED vs OFFERED RATE

0.05-core cap · msg/s



CPU CAP REMOVES THE BOTTLENECK

It's CPU-bound,

not a crypto-design limit.

Near saturation the whole pipeline converges — the profile gap shows in crypto work, not latency.

CONCLUSION

Adaptive crypto — measured, not just proposed

✓ **Formalised**

swarm graph + $F : S(t) \to P$

✓ **Built**

four modes incl. adaptive switching

✓ **Measured**

lightweight cheapest; adaptive allocates well

WHAT COMES NEXT

01 **Authenticated control plane**

signed bootstrap & updates

02 **Multi-peer validation**

churn · contention · async

03 **Hardware & post-quantum**

real platforms; hybrid PQ

Thank you — questions welcome.

github.com/mazzz3r/adaptive-swarm-crypto-thesis